GitHub OSS Ecosystem Overview (2016–2026)

Total Repositories	Total Contributors	Total Push Events	Total Pull Requests	Total Stars	Total Issues
repo_id 30,383	actor_id 1,789,163	total_pushes 1,240,714	total_prs 74,231	total_stars 80,626	total_issues 35,648

	year	pr_to_push_ratio	total_contributors	unique_repos	total_pushes	total_forks	total_issues	total_prs	total_stars
1.	2026	0.0061	25525	6200	189400	189	387	1148	932
2.	2025	0.0568	26101	6361	191706	1881	4878	10898	12082
3.	2024	0.0543	22736	6006	185087	1990	2632	10057	8572
4.	2023	0.0693	21741	5910	147437	2045	5502	10215	16593
5.	2022	0.0781	18958	5712	110213	1736	3348	8603	7090

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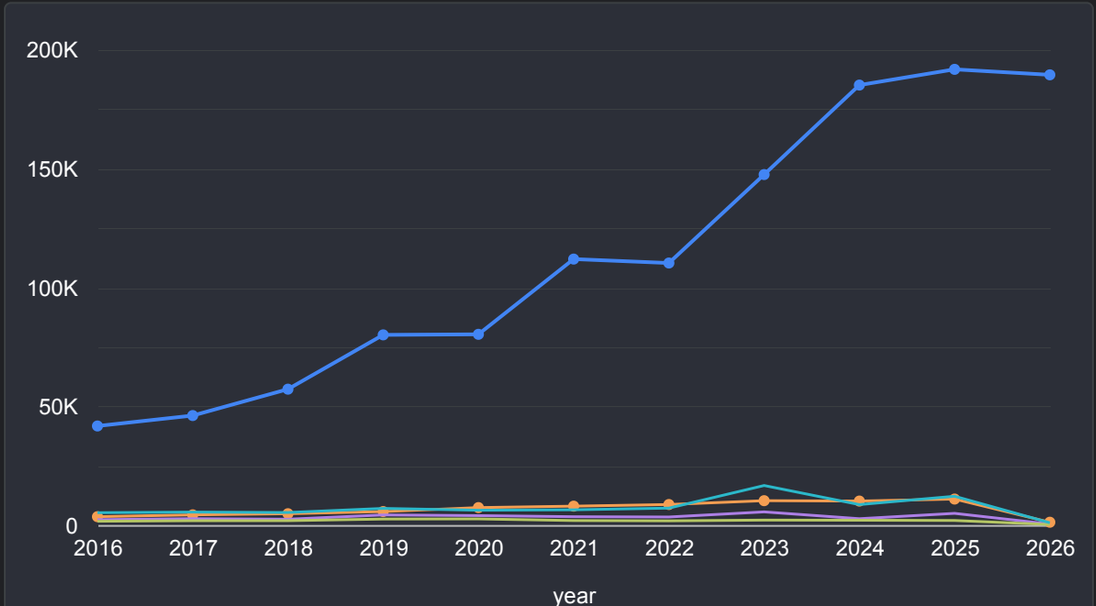
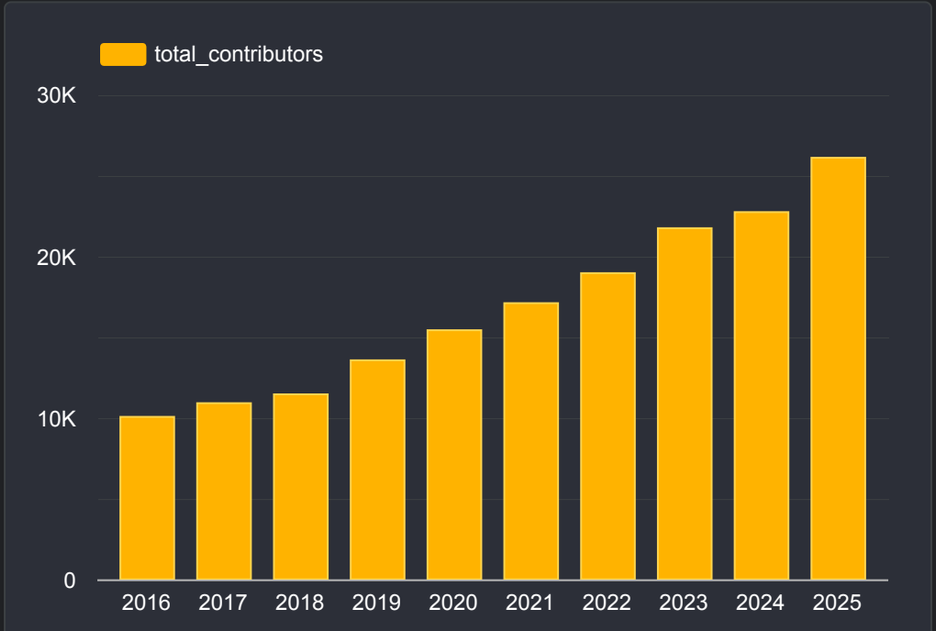
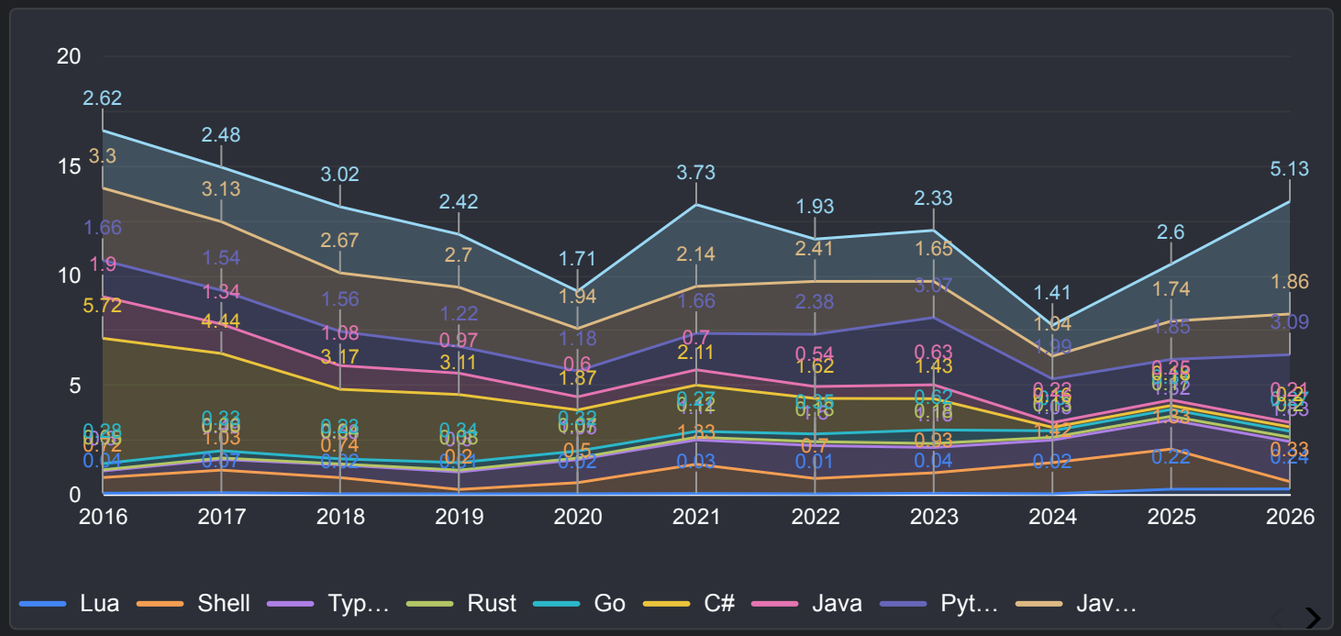
Ecosystem Growth by Year (2016–2026)





Programming Language Market Share (2016–2026)

Language Market Share Over Time



Total Languages

language_name
229

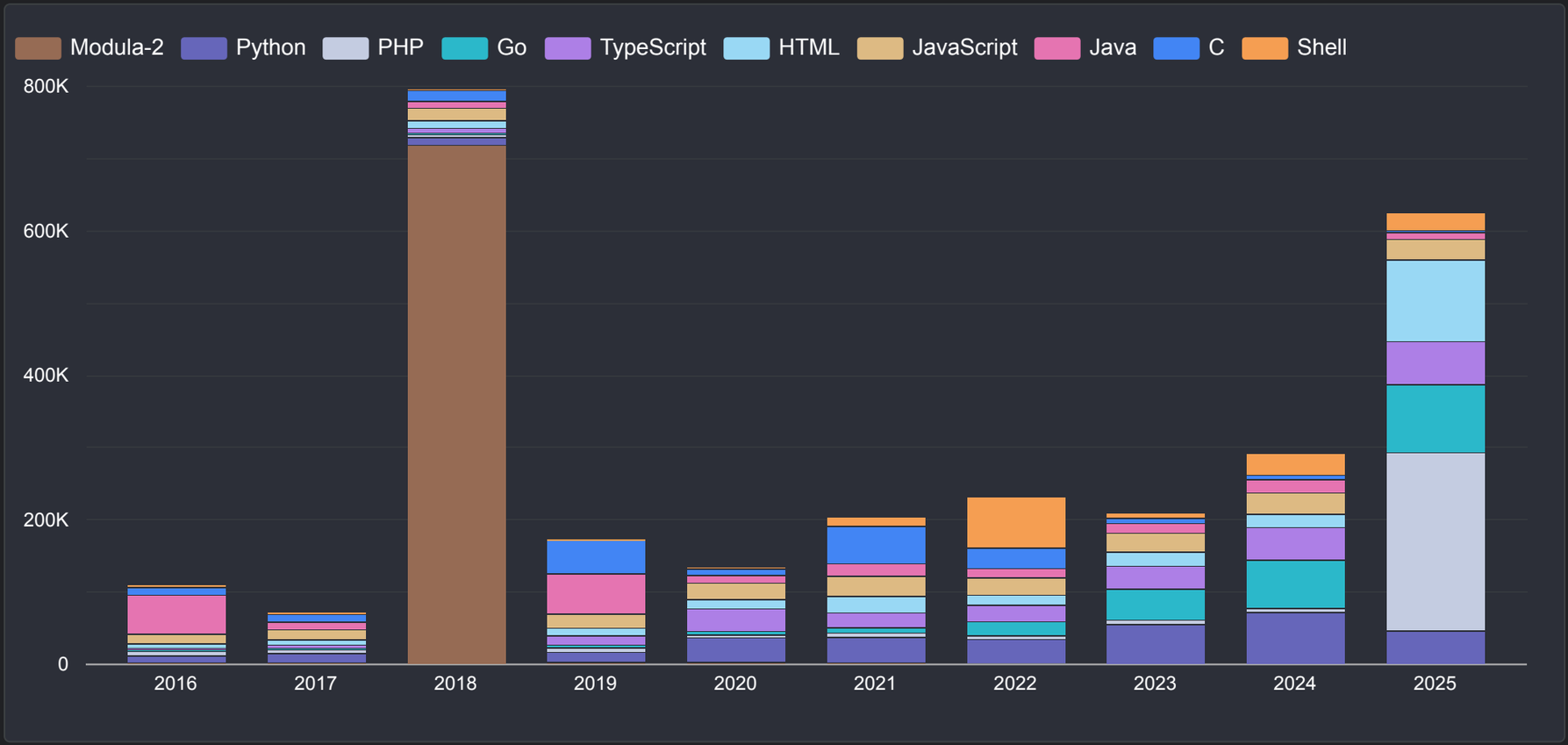
Description: Shows how each language's share of total GitHub push activity has changed over 10 years. Reveals the rise of Python/TypeScript and the relative decline of legacy languages. Relevance: Tracks the long-term shift in developer preferences, helping identify which languages are gaining or losing traction in the open-source ecosystem.

Language Detail Table

	language_name	year	market_share_pct	yoy_push_gro…	unique_repos	unique_contribut…
1.	HTML	2026	5.13	52.92	960	878
2.	Python	2026	3.09	29.09	895	1,105
3.	TypeScript	2026	1.83	6.77	779	1,379
4.	TypeScript	2024	1.03	32.17	729	1,355
5.	TypeScript	2025	1.32	29.09	697	1,591
6.	Python	2025	1.85	-6.51	674	1,009
7.	Python	2023	3.07	78.29	640	923
8.	Python	2024	1.99	-3.82	622	872
9.	JavaScript	2019	2.7	32.02	619	971
10.	JavaScript	2017	3.13	18.1	607	901
11.	TypeScript	2023	1.16	6.79	606	1,282

Description: Detailed drill-down table for exact numbers behind the charts.

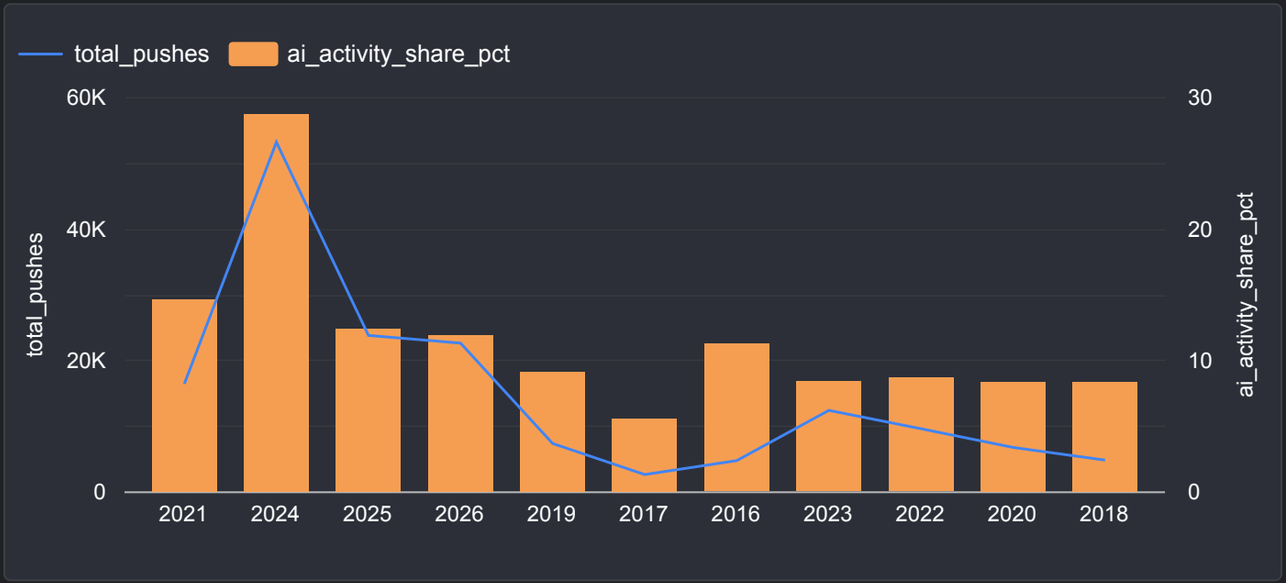
Total Commits by Language Per Year



Description: Shows absolute commit volume per language stacked by year. Unlike market share (relative), this shows whether total activity is growing even if share is flat.
Relevance: A language can lose market share while still growing in absolute terms, this chart surfaces that nuance.

AI/ML Boom — AI vs Non-AI Activity Over Time

AI/ML vs Non-AI Repository Activity



AI Repos

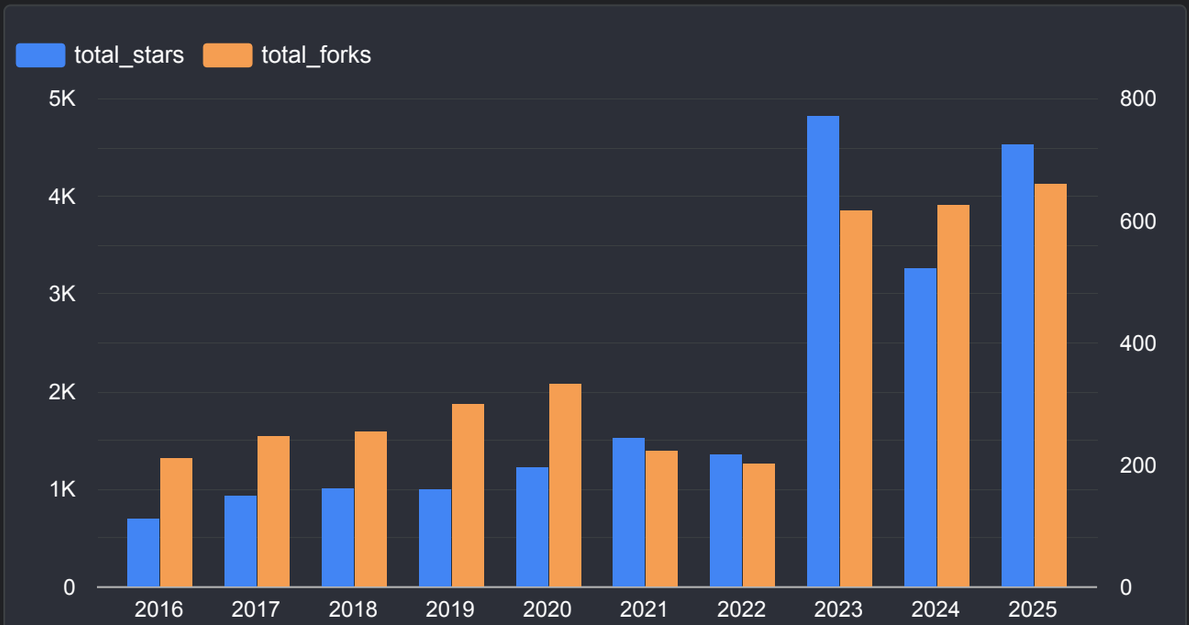
unique_repos
7,467

AI/ML Share of GitHub-2026

ai_activity_share_pct
11.9

Description: Bars show absolute push volume for AI/ML repos vs all others; the line shows AI/ML's growing share of total activity. The inflection point around 2022-2023 corresponds to the ChatGPT/LLM explosion.
Relevance Quantifies the AI/ML boom, not just that it happened, but exactly how much of GitHub's activity shifted toward AI projects.

AI/ML Repos: Stars & Forks Growth



AI/ML Share of GitHub-2026

unique_repos
838

AI/ML Share of GitHub-2026

total_pushes
22,534

Description: Shows the explosive growth in community engagement (stars) and derivative work (forks) for AI/ML repositories.
Relevance: Stars and forks are proxy measures for community interest and practical adoption, a spike here means developers are not just building AI tools but actively using and extending them.

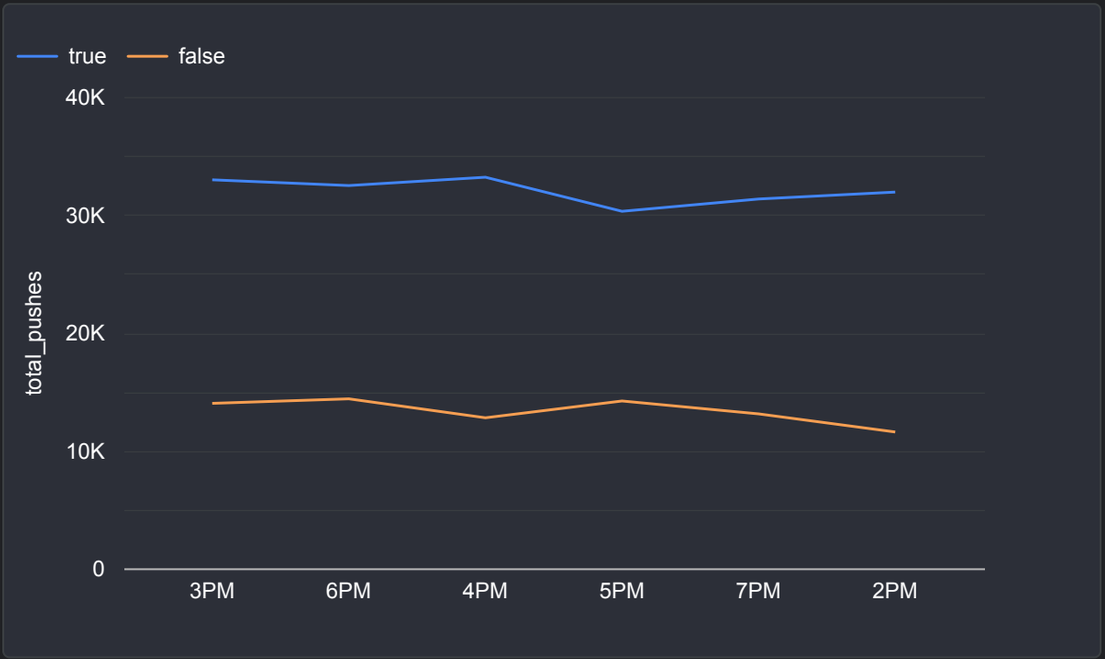
AI vs Non-AI Year-over-Year Comparison

	is_ai_ml_repo	year	unique_repos	total_pushes	total_stars	total_prs	ai_activity_share_pct
1.	true	2021	691	16,340	1,516	1,068	14.61
2.	true	2023	837	12,311	4,816	1,752	8.35
3.	true	2020	640	6,677	1,219	929	8.32
4.	true	2016	386	4,672	686	372	11.22
5.	true	2025	890	23,707	4,528	1,669	12.37
6.	true	2022	742	9,519	1,351	1,271	8.64
7.	true	2024	858	53,097	3,253	1,223	28.69
8.	true	2019	609	7,276	990	751	9.1
9.	true	2026	838	22,534	463	213	11.9
10.	true	2018	542	4,746	1,003	581	8.31
11.	true	2017	434	2,549	929	509	5.54
12.	false	2017	3,694	43,480	4,480	3,740	94.46
13.	false	2022	4,970	100,694	5,739	7,332	91.36
14.	false	2019	4,416	72,705	5,973	4,883	90.9
15.	false	2016	3,409	36,971	4,480	3,107	88.78
16.	false	2018	3,971	52,387	4,234	4,149	91.69
17.	false	2020	4,535	73,536	4,982	6,379	91.68
18.	false	2024	5,148	131,990	5,319	8,834	71.31
19.	false	2025	5,471	167,999	7,554	9,229	87.63
20.	false	2021	4,773	95,532	4,865	6,842	85.39
21.	false	2026	5,362	166,866	469	935	88.1
22.	false	2023	5,073	135,126	11,777	8,463	91.65

Description: Side-by-side numbers to compare AI/ML and non-AI/ML ecosystems across all metrics.

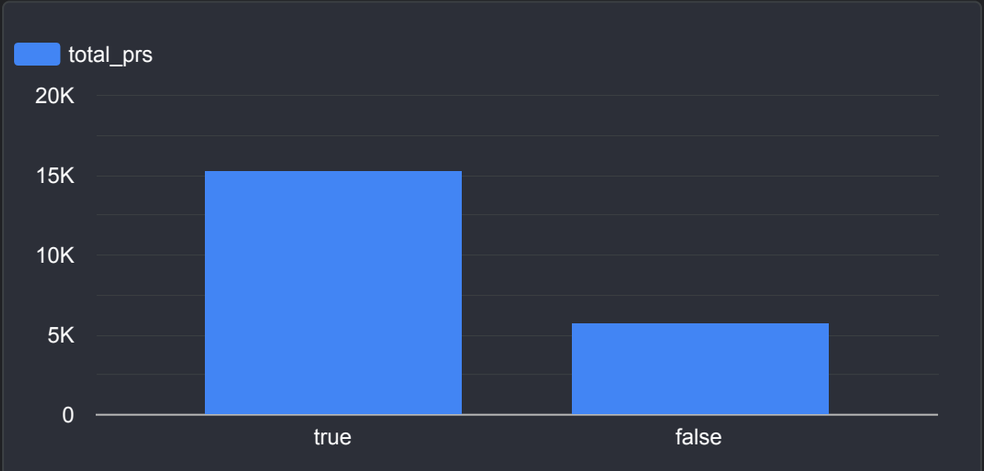
COVID-19 Impact on Developer Activity

Hourly Push Activity: Pre-COVID vs During COVID



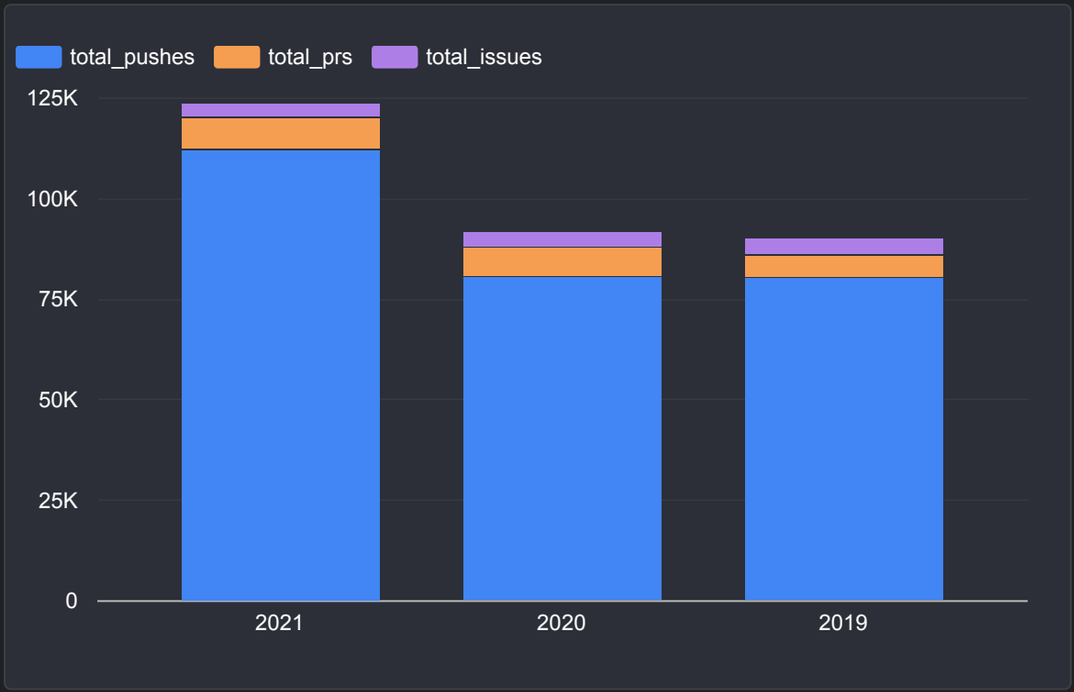
Description: Compares the hourly distribution of pushes before COVID vs during COVID.
Relevance: Reveals how remote work changed developer work patterns.

Pull Requests Raised: Pre-COVID vs During COVID



Description: Compares the total number of PRs raised during COVID (Mar 2020 - Jun 2021) vs the non-COVID months in 2019-2021. A spike during COVID indicates increased open-source collaboration during lockdowns.
Relevance: PRs are a proxy for collaborative development, a surge during COVID confirms that remote work drove more structured code review and open-source contribution.

Activity Type Breakdown by Year



Description: Shows total volume of pushes, PRs, and issues for 2019, 2020, 2021. Expect to see a surge in 2020 as developers pivoted to open-source during lockdowns.
Relevance: Measures the aggregate impact of COVID on developer productivity and open-source contribution.

Hour-by-Year Activity Heatmap

	hour	year	total_pushes
1.	6PM	2021	19,569
2.	4PM	2021	19,492
3.	2PM	2021	19,206
4.	3PM	2021	19,127
5.	7PM	2021	18,425
6.	5PM	2021	16,053
7.	6PM	2019	14,374
8.	5PM	2020	14,225
9.	5PM	2019	14,191

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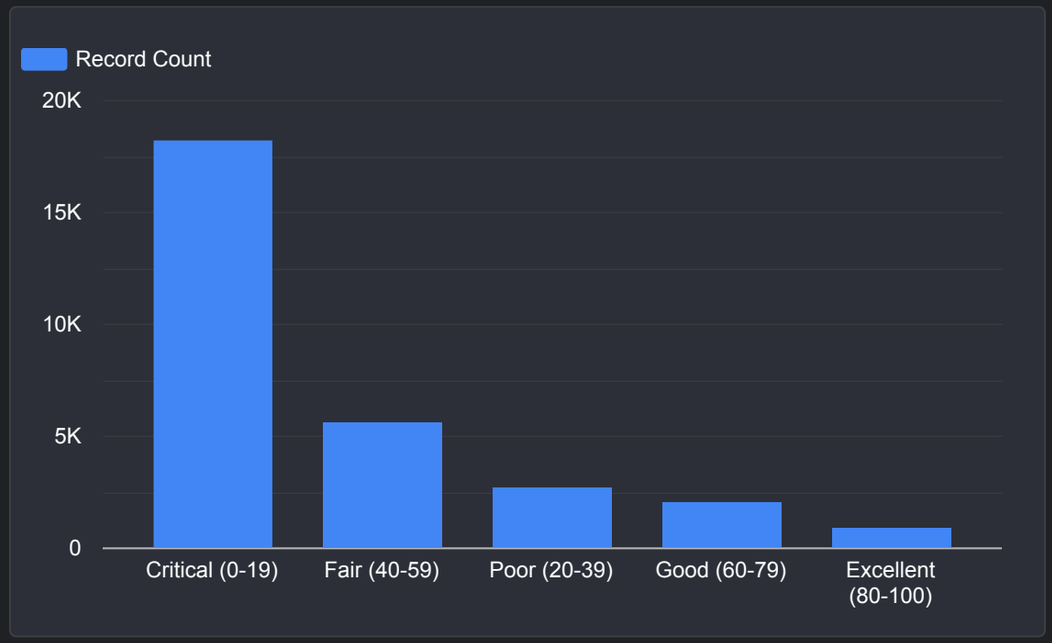
Description: Shows where push activity concentrates by hour for each year. During COVID, the "hotspot" should shift and spread.

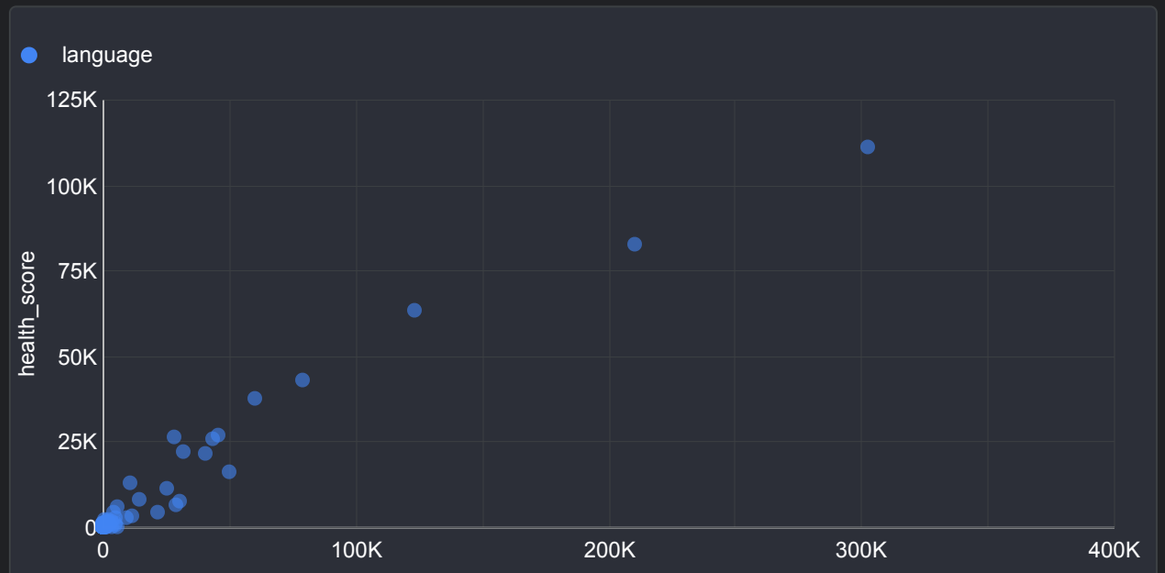
Repository Health Assessment

Health Score Distribution



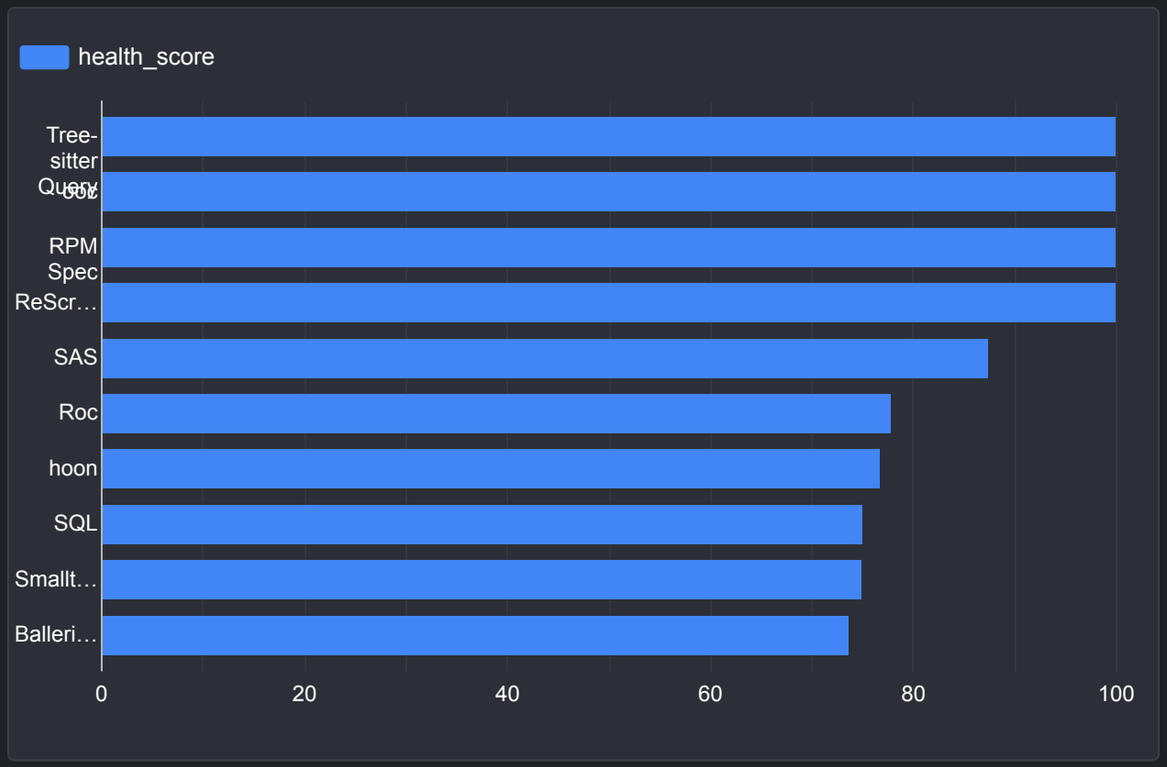
Description: Shows how many repositories fall into each health tier. Highlights the overall health of the ecosystem, a healthy ecosystem has most repos in the Good/Excellent range.
Relevance: Quickly identifies what proportion of the ecosystem is well-maintained vs at risk of deterioration.

PR Merge Time vs Health Score



Description: Reveals the relationship between PR responsiveness and overall health. Healthy repos cluster in the bottom-right (low merge time, high score).
Relevance: Identifies which metric most strongly drives health, useful for repo maintainers wanting to improve.

Average Health Score by Language



Description: Compares average repository health across languages.
Relevance: Helps identify which language ecosystems have the healthiest maintenance practices.

Top/Bottom Repos by Health Score

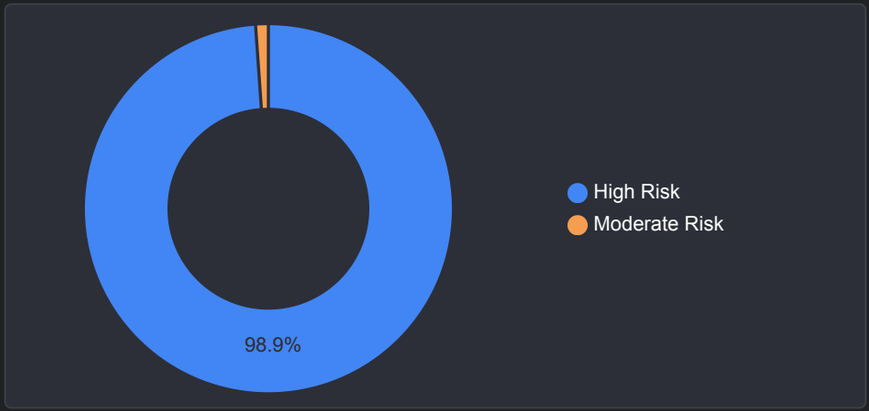
	language	repo_name	is_ai_ml_repo	health_s…	avg_pr_merge_hrs	avg_issue_close_…	merge_rate	total_stars
1.	C	TulipCharts/t…	false	100	7,718.13	null	9	1
2.	Vue	feathersjs-e…	false	100	6.16	610.03	3	0
3.	Scala	lagom/lagom	false	100	38.49	2,702.73	4	2
4.	C#	mRemoteN…	false	100	0.84	1,382.73	4	8
5.	PHP	PHP-FFMpe…	false	100	335.52	204.28	5	1
6.	TypeScript	mei23/misskey	false	100	8.07	15.71	10	0
7.	Dart	verhneudine…	false	100	100.81	null	7	0
8.	Visual Basi…	IDEMSInter…	false	100	346.99	521.32	4	0
9.	Go	fnproject/fn	true	100	9.2	null	4	3

Drill-down table to identify specific top-performing and at-risk repositories.

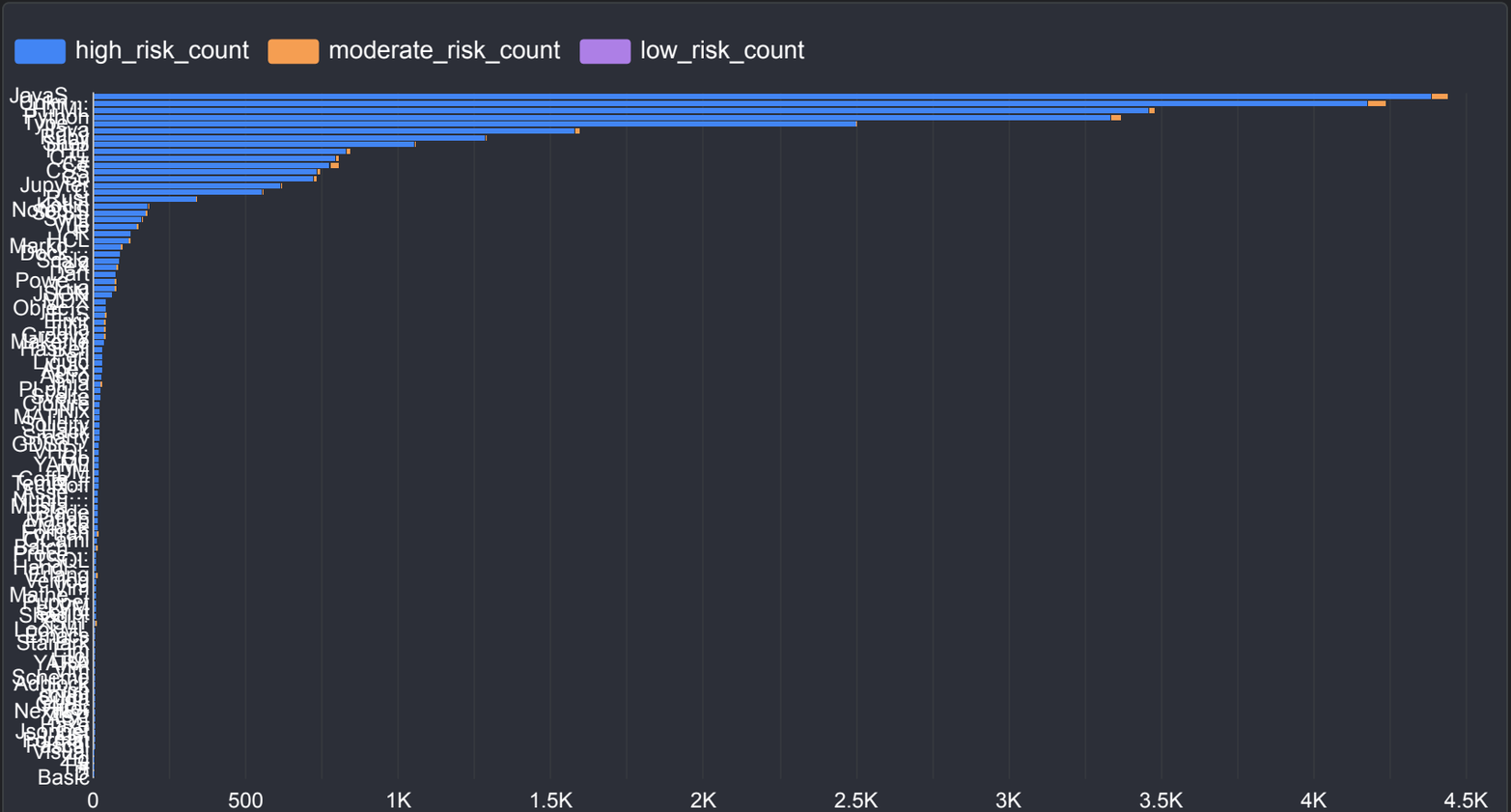
Repository Abandonment Risk Prediction

Description: Shows the proportion of repositories in each risk tier. A large red slice indicates systemic ecosystem health issues.
Relevance: Provides an instant read on what fraction of the ecosystem is at risk of being abandoned.

Overall Risk Distribution



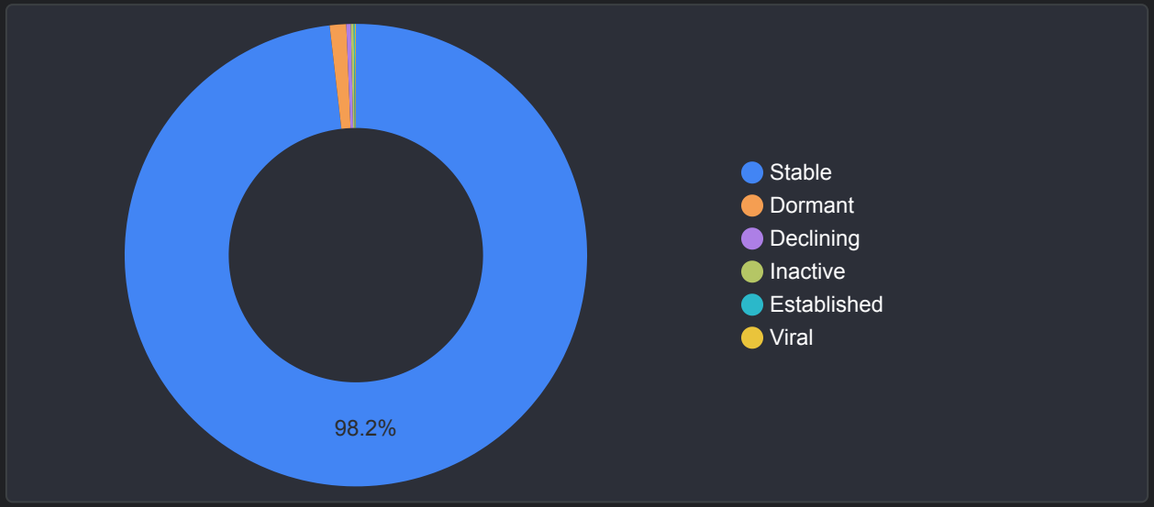
Abandonment Risk by Language



Description: Shows which languages have the highest proportion of at-risk repos. Languages with small communities or declining popularity tend to have higher risk.
Relevance: Identifies language ecosystems where project sustainability is a concern.

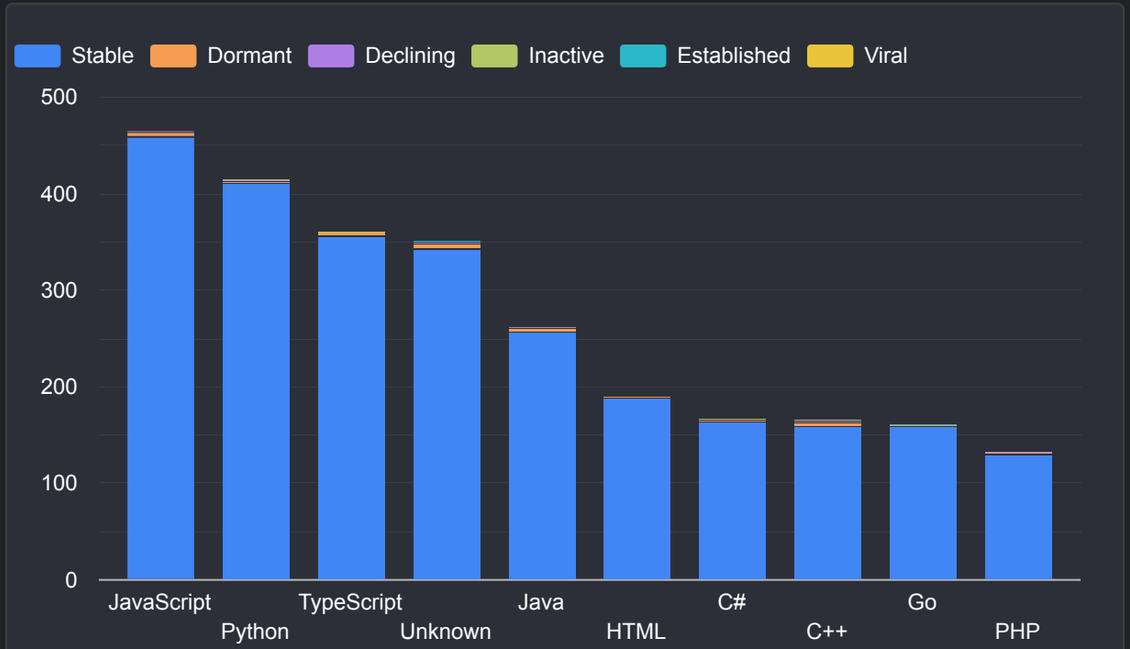
Repository Growth Trajectories

Trajectory Distribution



Description: Shows what proportion of repos fall into each growth trajectory. A healthy ecosystem has a mix of Viral + Established and relatively few Dormant/Inactive.
Relevance: Gives a snapshot of ecosystem vitality, are repos growing, stagnating, or dying?

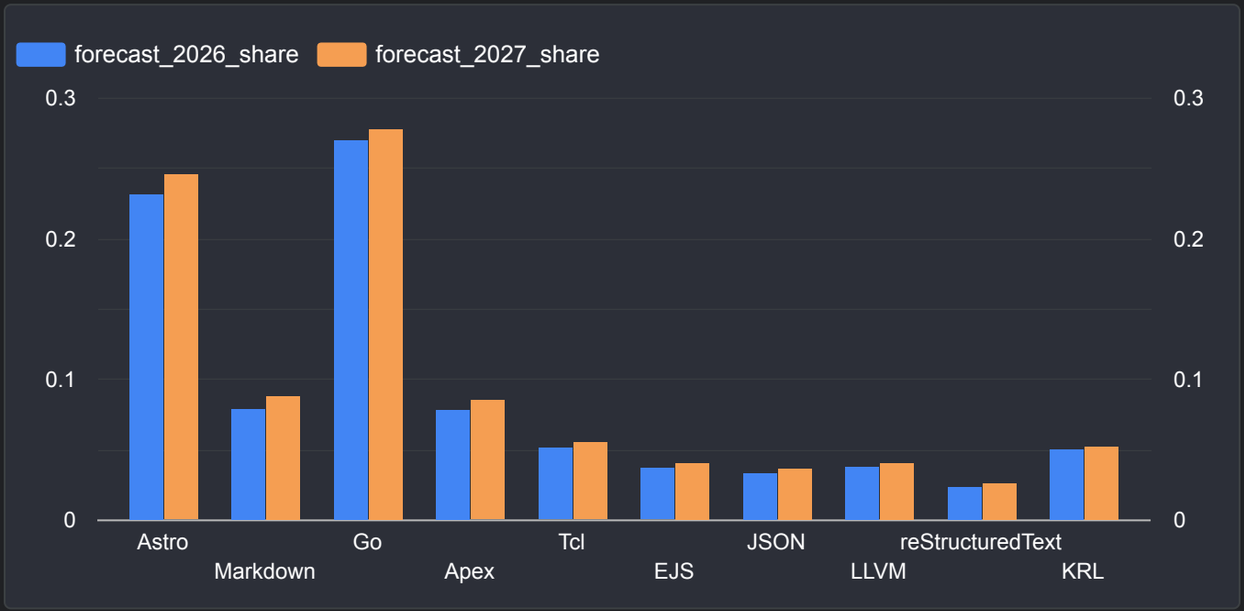
Trajectory by Language



Description: Shows each language's distribution across growth trajectories. A language dominated by "Declining" clusters may be losing relevance.
Relevance: Connects growth trajectory analysis back to the language-level story.

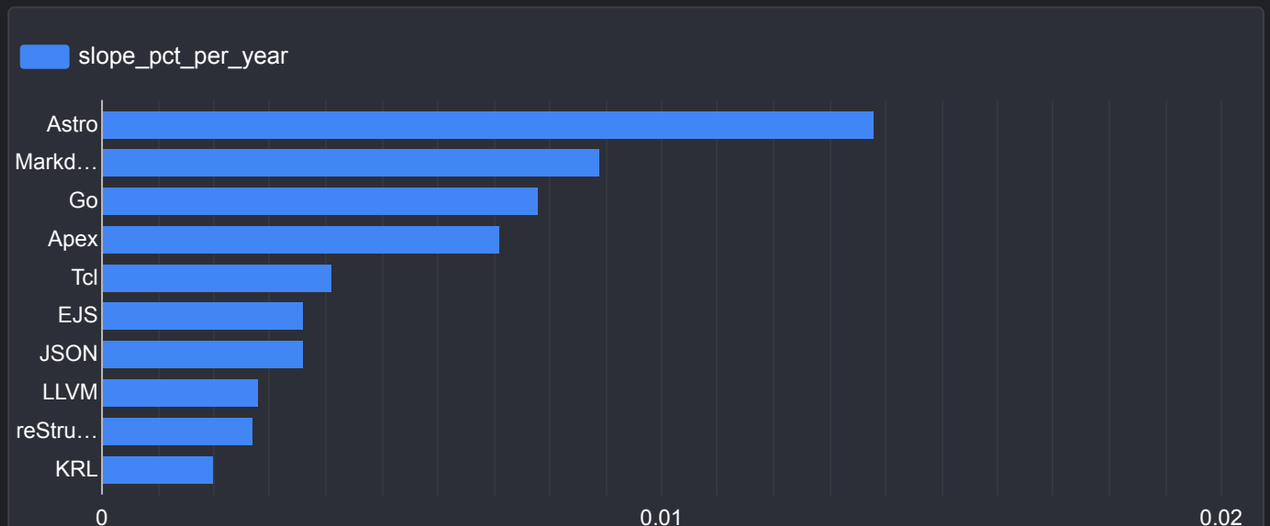
Language Market Share Forecast

Forecast: 2026 vs 2027 Market Share



Description: Side-by-side bars showing predicted market share for each language in 2026 and 2027. Languages with growing bars are gaining traction; shrinking bars are losing ground.
Relevance: Provides forward-looking intelligence for technology investment decisions and hiring strategies.

Growth Slope (pp/year)



Description: Shows the annual rate of market share change per language. A slope of +0.5 means the language gains 0.5 percentage points of market share each year.
Relevance: Separates "growing fast" from "growing slowly", a language with 2% share but +0.5 slope is more dynamic than one with 15% share and +0.1 slope.